

Rashi Jaiswal, Ph.D

Senior Software Engineer (AI/ML)

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PROFESSIONAL SUMMARY

Experienced researcher specializing in Data science with a Ph.D. in Computer Science. Expertise in machine learning, time series analysis, computer vision, anomaly detection, and NLP, with a proven track record of implementing cutting-edge solutions in both industry and academia. Strong leadership skills with experience managing teams and leading projects. Published researcher with multiple patents and journal articles in AI, deep learning, and spatial-temporal data mining. Interested to contribute with AI based solutions in Healthcare, Education, Manufacturing, finance industries.

PROFESSIONAL EXPERIENCE

Senior Software Engineer (AI/ML)

DIMAAG-AI | Bangalore, India

March 2023 – Present

- Developed unsupervised activity clustering using spatial-temporal feature embeddings for real-time action label prediction with improved accuracy to do sequence monitoring.
- Conducted time series forecasting using RNN, LSTM, GRU, and regression models (Ridge, Lasso, Elastic Net) achieving 90-95% accuracy.
- Implemented generative models and LLM for document parsing and NLP tasks.
- Led the development of a multi-camera, multi-object tracking system for person reidentification using spatial-temporal features.
- Worked on anomaly detection in CT image data using models like Auto-Encoders, VAE, GAN, and Vision Transformers, achieving 98% accuracy.
- Fine-tuned models such as ResNet, EfficientNet, MobileNet, and others for computer vision tasks (classification, detection, segmentation, etc.).
- Supervised a team of 8-9 members and managed end-to-end AI pipelines from data preprocessing to model deployment.

Assistant Professor

GLA University | Mathura, U.P., India

October 2022 – February 2023

- Taught and mentored students in Machine Learning, Deep Learning, and Data Structures.
- Contributed to the NAAC Accreditation Team and handled academic administration tasks.

Doctoral Researcher

ICT Research Lab, University of Lucknow | Lucknow, U.P., India

March 2019 – August 2022

- Developed an automated prediction system for time series data, improving human activity recognition in defense and healthcare.
- Created TConvRec, a model enhancing pattern learning for time series data.
- Proposed novel approaches for synthetic temporal data generation to address data scarcity issues.
- Study of Parkinson's Disease progression with Time series by analyzing the patient's body movement pattern over time.

EDUCATION

- **Ph.D. in Computer Science**
University of Lucknow | 2022
Thesis: *Study on Data Mining Techniques for Temporal Data Learning*
- **M.C.A. (Master of Computer Applications)**
University of Lucknow | 2018 | 82.52%
- **B.C.A. (Bachelor of Computer Applications)**
University of Lucknow | 2015 | 72.08%

TECHNICAL SKILLS

- **Data Science Tools:** Numpy, Pandas, Scikit-learn, TensorFlow, Keras, PyTorch, CUDA
- **Machine Learning & AI:** CNNs, LSTM, ResNet, EfficientNet, VAE, GAN, Transformers
- **Programming:** Python, Java, C, C++, JavaScript, PHP
- **NLP:** BERT, GPT, LLMs, RAG
- **Database:** MySQL, MongoDB, Oracle, SQL Server
- **Cloud Platforms:** AWS, GCP
- **Development:** Flask, FastAPI, REST API, MVC architecture

PUBLICATIONS & PATENTS

Patents Contributions:

- *Automated Prediction System for Temporal Data*, 2022, Application No.: 202211002659.
- *System for Prediction Modeling for Synthetic Time Series Metadata Fusion*, 2022. Application No. : 202211018390.

Publications Contributions:

- Singh, B., & Jaiswal, R. (2022). *Stride Window Approach with Anomaly Detection for Probability Risk Assessment*. Springer Nature.

- Singh, B., & Jaiswal, R. (2022). *Automation of Prediction System for Temporal Data*. International Journal of Information Technology, Springer.
- Jaiswal, R., & Singh, B. (2022). *Financial Fraud Prevention with Synthetic Data Generation using GAN*. Aryabhata Journal of Mathematics and Informatics.
- Singh, B., & Jaiswal, R. (2022). *TConvRec: Temporal Convolutional-Recurrent Fusion Model with Additional Pattern Learning*. International Journal of Information Technology, Springer.
- Jaiswal, R. (2023). *Stride Window Approach in Multiclass Classification for Temporal Sensor Data Augmentation*. Aryabhata Journal of Mathematics and Informatics.
- Jaiswal, R., & Singh, B. (2023). *A Review on Temporal Data Mining*. Aryabhata Journal of Mathematics and Informatics.
- Singh, B., & Jaiswal, R. (2021). *Automation of Prediction Method for Supervised Learning*. IEEE Confluence.
- Singh, B., & Jaiswal, R. (2021). *Impact of Hybridization of Deep Learning Models for Temporal Data Learning*. IEEE UPCON.
- Jaiswal, R., & Singh, B. (2021). *Comparative Study of Loss Functions for Deep Neural Networks in Time Series Analysis*. Springer BigDML.
- Jaiswal, R., & Singh, B. (2022). *Hybrid Convolutional Recurrent (CNN-GRU) Model for Stock Price Prediction*. IEEE CSNT-2022.
- Singh, B., & Jaiswal, R. (2022). *Study on Classifiers for Temporal Data*. IEEE CSNT-2022.
- Jaiswal, R. (2024). *Prediction Model for Synthetic Time Series Metadata Fusion*. IEEE Confluence-2024.

CERTIFICATIONS

- CloT-2024, AWS Workshop on cloud computing in IoT | Bennett University, India, 2024
- Satellite Remote Sensing of Atmosphere | ISRO, 2022
- Applications of Machine Learning in Signal, Image, and Computer Vision | IEEE, 2021
- Emerging IT Technologies | MNNIT Allahabad, 2020

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